

**Peer influence effects on the Bandit Task: A four-armed bandit experiment regarding peer information and its impacts on exploration and exploitation.**

Anastassia Olijnyk, Brayden Tardiff, Emma Grindstaff and Nina Wolff

Department of Psychology, University of Wisconsin-Madison

PSYCH 505: Learning and Decision-Making – Spring 2026

Discussion Section 301

May 3rd, 2026

**Introduction** (Introduction section written by: Anastassia Olijnyk, Hypothesis section written by: Emma Grindstaff, edited by: Anastassia Olijnyk)

The bandit task is implemented in the study of decision making and learning, particularly in the context of the credit assignment problem. At its core, this task aims to determine how individuals attribute outcomes to specific action. The goal is to maximize the cumulative reward, with each trial requiring a choice to be made between multiple options (Toyokawa et al., 2017). This is followed by observation of the resulting outcome. However, each option has an unknown reward probability. The principle tension of the bandit task is balancing the acts of exploration, seeking out new information, and exploitation, only selecting options with a known high reward. Often these tasks are made and replicated in an asocial environment, but it is known that many forms of learning and credit assignment can be formed in social settings as well (Toyokawa et al., 2017).

Previous research has primarily focused on credit assignment with comparison between asocial and social environments. It was found that an asocial bandit task environment did not predict the social environment credit assignment results (Danwitz & Helversen, 2025). However, within the social environment, most social information was found to be bonus information to the already observed information and knowledge formed from individual choice making (Danwitz & Helversen, 2025). This was a common tactic when the individual making the choice was faced with uncertainty. As well, the tension between exploitation and exploration was observed under social influence conditions, finding that participants were more likely to copy explorative agents than exploitative ones (Toyokawa et al., 2017).

Furthermore, the neurological bases of decision making and peer influence have been studied as well. Evidence of brain structures, specifically the striatum, modulating associated values and the updating of values is a common theme. Within the striatum there are discriminatory pathways for decision making as well. Using the neurotransmitter, dopamine, the dorsal and ventral pathways are split, with the dorsal providing outcome signals and the ventral pathways rendering action signals (Kim et al.,

2009). As well, it's suggested that the occipital-parietal network links perceptual and emotional processing to social conformity as well as peer pressure (Berns et al., 2005). These neural implementations provide a framework for the actor-critic model seen in the decision making and learning process of humans (Liakoni et al., 2022).

Although research has indicated that social influence plays a role in credit assignment, little is known regarding the timing of this influence during the bandit task. This study aims to test peer influence and its impact on individual credit assignment across varying temporal conditions. Specifically, peer responses will be shared within two modalities; immediate feedback and interval-based feedback. With the manipulation of these conditions, the relationship between the frequency and timing of social influence in the bandit task will be explored.

Specifically, we predict that during the social information conditions (intermittent and continuous blocks), participants will shift toward socially guided exploration, preferentially selecting options chosen by a higher percentage of their peers, relative to the baseline condition (private block), in which no peer information is provided. Similarly, a social decision making task conducted by Adrian et al. (2019) hypothesized that participants would choose either an egocentric, decisions based on their own outcome, or joint strategy, taking into account their partner's outcomes as much as their own. Their "ice-fishing" game revealed that nearly all participants, when playing against a "partner", utilized a joint strategy. We expect our own study to yield a similar outcome, which will be shown by a significant main effect of block type on social conformity score. The continuous and interval blocks will have higher social conformity scores, as shown by post hoc comparison.

Furthermore, we predict that this shift in behavior will be reflected such that participants in the social information blocks, specifically the continuous block, will show a lower exploration rate and demonstrate greater reliance on social information. This pattern will manifest as a significant main effect of block type on switching behavior. Post hoc tests will show a lower total number of switches for these blocks, indicating heightened conformity-driven exploitation when peer behavior is visible.

Following previous research findings regarding collective inputs in individual decision making, it is predicted that peer influence and the addition of collective choice feedback, will create a greater impact for conformity in the bandit task (Charpentier et al., 2015). It is hypothesized that participants will exhibit a higher percentage of socially guided exploration overall when peer choice information is available, with conformity heightened within the continuous block compared to the interval block or when peer information is absent.

### **Methods** (Written by: Nina Wolff)

An online experiment was run with 43 college-aged participants. Participants completed a web-based four-armed bandit task. On each trial, participants chose between four colored squares that each had different probabilities of reward. The reward value was 10 points for each square. The probability of receiving a reward after clicking each square was either 20%, 40%, 60%, and 80%. The assignment of reward probability to square color was shuffled in each block. Participants were told immediately after they clicked a square whether or not they received points. The goal for participants was to get the highest amount of accumulated points at the end of the block compared to past participants.

Each participant was instructed to complete three blocks in a randomized order with 30 trials in each block. In the private block, participants were only able to see how many points they were accumulating throughout the task, they could not see how any other participants performed. At the end of the block, participants could see how they performed on a points based leaderboard of past participants. In the continuous block, a leaderboard was visible throughout the entire block. It showed how participants were currently performing based on their accumulated points compared to past participants at that specific trial in the block. It also showed the percentage of past participants that had chosen each color square, specifically highlighting the most popular square color choice at that trial. In the intermittent block, that same leaderboard was visible, but only every five trials.

To examine the effect of peer influence on decision making, two variables were examined: the total number of switches (defined as choosing a different square than on the previous trial) and a social conformity score (the proportion of trials where the participant's choice matches the highest-ranked option on the leaderboard) for each block.

## **Results** (Written by: Brayden Tardiff)

A total of 2,970 trials from participants ( $N = 43$ ) were completed across three different block conditions: Block 1 (Private and no peer information available,  $n = 990$  trials), Block 2 (Continuous: repeated peer choice feedback,  $n = 960$  trials), and Block 3 (Intermittent feedback: feedback available after every 5 trials,  $n = 1,020$  trials). Some participants did not complete all three blocks but their data was included in data analysis. Two dependent variables were examined: switching behavior ( $Is\_Switch$  : 0 = stayed with prior choice, 1 = switched), operationalizing exploration vs exploitation, and social conformity score, which is the proportion of choices matching the majority peer choice, operationalizing peer influence. All analyses used two-tailed tests with  $\alpha = .05$ .

### **Switching Behavior (Exploration vs. Exploitation)** (Written By Brayden Tardiff)

Descriptive statistics revealed a clear order of switching rates between all conditions (see table 1). The intermittent block produced the highest mean switch rate ( $M = 0.422$ ,  $SD = 0.494$ , 95% CI [0.391, 0.452]), followed by the Private block ( $M = 0.363$ ,  $SD = 0.481$ , 95% CI [0.333, 0.393]), with the Continuous block exhibiting the lowest switching ( $M = 0.359$ ,  $SD = 0.480$ , 95% CI [0.329, 0.390]). Contrary to the hypothesis that the intermittent and continuous blocks would both show reduced switching relative to private baseline, only the continuous condition showed a small directional decrease while the intermittent condition showed a substantial increase.

A one-way between-subjects ANOVA revealed a significant main effect of block condition on switching behavior,  $F(2, 2967) = 5.22$ ,  $p = .005$ ,  $\eta^2 = .004$ . Although the effect size is small by conventional benchmarks, the result confirms that block condition reliably influenced exploratory responding.

Post-hoc independent samples t-tests with Bonferroni correction showed that switching was significantly higher in the intermittent condition than in the continuous condition,  $t(1978) = -2.84$ ,  $p = .005$ ,  $d = -0.128$ . No other pairwise comparisons remained significant after correction.

These results indicate that intermittent peer feedback, rather than suppressing exploration, actually elevated switching relative to the no-information baseline and the continuous feedback condition. This is partially inconsistent with the initial hypothesis that social conditions would reduce exploration. One interpretation, consistent with reinforcement learning theory, is that the intermittent social information introduces uncertainty about which option is optimal, thereby promoting renewed exploration, analogous to partial reinforcement schedules that increase response variability. Alternatively, participants may have used intermittent peer updates as explicit cues to re-explore when the visible peers chose a different option, a strategy consistent with directed exploration models (Wilson et al., 2014)

### **Social Conformity** (Written by: Brayden Tardiff)

Social conformity scores (proportion of trials on which a participant chose the same option as the majority of visible peers) were examined across the three block conditions. Descriptive statistics are available in table 1. The continuous block showed the numerically highest conformity ( $M = 0.336$ ,  $SD = 0.473$ , 95% CI [0.307, 0.366]) followed by the private block ( $M = 0.288$ ,  $SD = 0.453$ , 95% CI [0.260, 0.316]), with the intermittent block showing lowest conformity ( $M = 0.313$ ,  $SD = 0.464$ , 95% CI [0.284, 0.341]).

Critically, a one way ANOVA revealed no significant main effect of block condition on social conformity,  $F(2,2967) = 2.68$ ,  $p = .069$ ,  $\eta^2 = .002$ . Post-hoc Bonferroni-corrected t-tests confirmed no pairwise differences.

A chi-square test of independence examining whether the proportion of trials with any social conformity (conformity > 0 vs conformity = 0) differed by block condition was also non-significant,  $\chi^2(2, N = 2970) = 5.36$ ,  $p = .069$ , Cramer's  $V = .042$ .

These results do not support the hypothesis that social conditions would produce higher conformity scores. Conformity scores numerically trended upward in social

conditions, with the Continuous block showing the highest mean conformity and both social blocks exceeding the private baseline, though these differences did not reach statistical significance.

The sample size of 43 participants may have limited power to detect conformity effects; the effect size observed ( $\eta^2 = .002$ ) is small, and a larger sample would be needed to draw firm conclusions.

**Table 1** (Created by: Brayden Tardiff)

**Descriptive Statistics and Inferential Results for Switching Rate and Social Conformity Across Block Conditions**

| Measure           | Block             | M (SD)        | 95% CI         | Overall statistic                            |
|-------------------|-------------------|---------------|----------------|----------------------------------------------|
| Switching rate    | Private (B1)      | 0.363 (0.481) | [0.333, 0.393] |                                              |
|                   | Continuous (B2)   | 0.359 (0.480) | [0.329, 0.390] |                                              |
|                   | Intermittent (B3) | 0.422 (0.494) | [0.391, 0.452] | F(2,2967)=5.22<br>, p=.005,<br>$\eta^2=.004$ |
| Social conformity | Private (B1)      | 0.288 (0.453) | [0.260, 0.316] |                                              |
|                   | Continuous (B2)   | 0.336 (0.473) | [0.307, 0.366] |                                              |
|                   | Intermittent (B3) | 0.313 (0.464) | [0.284, 0.341] | F(2,2967)=2.68<br>, p=.069,<br>$\eta^2=.002$ |

## **Discussion** (Written By Brayden Tardiff)

Our hypothesis stated that participants in social conditions (Intermittent and Continuous blocks) would exhibit different decision making strategies, specifically reduced switching and increased social conformity compared to the private block. Our hypothesis received mixed support.

For switching behavior, the prediction was partially reversed: rather than both social conditions showing less exploration, the intermittent block produced more switching than the Continuous condition . Only the continuous block showed a (non-significant) directional decrease in switching, which is more consistent with the hypothesis.

The finding that continuous peer feedback reduces exploration while intermittent feedback increases it maps onto theoretical accounts of social learning under uncertainty. From a computational perspective, having constant access to peer choices may lower the need to keep exploring because participants can use the social information as a cue about which option is best. In this way, some of the learning is supported by the social environment instead of only trial-and-error learning. This is consistent with Bayesian accounts of social influence (Behrens et al., 2008), which suggest that dependable social cues can reduce how much people rely on their own learning. On the other hand, intermittent peer feedback may create uncertainty about whether a previously observed peer preference is still correct, which could then encourage participants to start exploring again between updates. This pattern is very similar to volatility-driven exploration described in adaptive learning research (Behrens et al., 2007).

The lack of significant conformity differences across blocks is interesting when compared to the switching results. A possible explanation is that the social conformity variable may reflect agreement with majority peer choices after the fact rather than intentionally copying during the task. In other words, participants in the Intermittent block who explored more may have independently found better options that also happened to match what peers selected. Under this view, switching and conformity may not be fully separate processes, but instead different outcomes of the same exploratory behavior. Future studies could look at this more closely by testing whether switches are

directed toward previously observed peer choices (imitative exploration) or toward completely new options (independent exploration).

### **Limitations** (Written by: Brayden Tardiff)

Several limitations need to be considered when looking at our experiment. One that may go unnoticed, is that each participant completed multiple trials, meaning that responses from the same individual would likely be related rather than completely independent. This could potentially reduce the accuracy of analyses that assume every observation is separate. Secondly, our social conformity score may not perfectly represent true imitation score, because participants could match the majority peer choice by independently choosing the most rewarding option. Another struggle that the group had, was when we sent out our experiment, there was a cluster of participants that only did one of three sections, which limits the strength of within-subject comparisons. Lastly, our sample size was 43, which may have reduced statistical power, particularly for detecting smaller conformity effects. I believe that for future studies having a larger sample size we would be able to more clearly see the effects of social conformity.

## References

- Adrian, J. A., Siddharth, S., Baquar, S. Z. A., Jung, T.-P., & Deák, G. (2019). Decision-making in a social multi-armed bandit task: Behavior, electrophysiology and pupillometry. arXiv. <https://doi.org/10.48550/arXiv.1905.07474>
- Behrens, T. E. J., Hunt, L. T., Woolrich, M. W., & Rushworth, M. F. S. (2008). Associative learning of social value. *Nature*, 456(7219), 245–249. <https://doi.org/10.1038/nature07538>
- Behrens, T. E. J., Woolrich, M. W., Walton, M. E., & Rushworth, M. F. S. (2007). Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9), 1214–1221. <https://doi.org/10.1038/nn1954>
- Berns, G. S., Chappelow, J., Zink, C. F., Pagnoni, G., Martin-Skurski, M. E., & Richards, J. (2005). Neurobiological correlates of social conformity and independence during mental rotation. *Biological psychiatry*, 58(3), 245–253. <https://doi.org/10.1016/j.biopsych.2005.04.012>
- Charpentier, C. J., Moutsiana, C., Garrett, N., & Sharot, T. (2014). The brain's temporal dynamics from a collective decision to individual action. *The Journal of neuroscience : the official journal of the Society for Neuroscience*, 34(17), 5816–5823. <https://doi.org/10.1523/JNEUROSCI.4107-13.2014>
- Chen, J., Wu, Y., Tong, G., Guan, X., & Zhou, X. (2012). ERP correlates of social conformity in a line judgment task. *BMC neuroscience*, 13, 43. <https://doi.org/10.1186/1471-2202-13-43>
- Danwitz, L., & von Helversen, B. (2025). Observational learning of Exploration-exploitation strategies in bandit tasks. *Cognition*, 259, 106124.

<https://doi.org/10.1016/j.cognition.2025.106124>

Kim, H., Sul, J. H., Huh, N., Lee, D., & Jung, M. W. (2009). Role of striatum in updating values of chosen actions. *The Journal of neuroscience : the official journal of the Society for Neuroscience*, 29(47), 14701–14712.

<https://doi.org/10.1523/JNEUROSCI.2728-09.2009>

Liakoni, V., Lehmann, M. P., Modirshanechi, A., Brea, J., Lutti, A., Gerstner, W., & Preuschoff, K. (2022). Brain signals of a Surprise-Actor-Critic model: Evidence for multiple learning modules in human decision making. *NeuroImage*, 246, 118780.

<https://doi.org/10.1016/j.neuroimage.2021.118780>

Park, S. A., Goïame, S., O'Connor, D. A., & Dreher, J. C. (2017). Integration of individual and social information for decision-making in groups of different sizes. *PLoS biology*, 15(6), e2001958. <https://doi.org/10.1371/journal.pbio.2001958>

Toyokawa, W., Saito, Y., & Kameda, T. (2017). Individual differences in learning behaviors in humans: asocial exploration tendency does not predict reliance on social learning. *Evolution and Human Behavior*, 38(3), 325-333.

<https://doi-org.ezproxy.library.wisc.edu/10.1016/j.evolhumbehav.2016.11.001>

Wilson, R. C., Geana, A., White, J. M., Ludvig, E. A., & Cohen, J. D. (2014). Humans use directed and random exploration to solve the explore–exploit dilemma. *Journal of Experimental Psychology: General*, 143(6), 2074–2081.

<https://doi.org/10.1037/a0038199>

### **Contributor Block**

Programming: Nina W.(primary), Anastassia O.(debugging)

Presentation: Anastassia O.(intro), Brayden T. (results), Emma G. (hypothesis) and Nina W.(web app)

Introduction: Anastassia O. (Primary), Emma G (hypothesis).

Methods: Nina W.

Results and Discussion: Brayden T.

References: Anastassia O., Brayden T